

DenoMAE2.0: Improving Denoising Masked Autoencoders by Classifying Local Patches for Automatic Modulation Classification

Atik Faysal, *Member, IEEE*, Mohammad Rostami, *Student Member, IEEE*, Taha Boushine, *Student Member, IEEE*, Reihaneh Gh. Roshan, *Student Member, IEEE*, Huaxia Wang, *Member, IEEE*, Nikhil Muralidhar, *Member, IEEE*, Yu-Dong Yao, *Fellow, IEEE*

Abstract—We introduce DenoMAE2.0, an enhanced denoising masked autoencoder designed to significantly improve representation learning for automatic modulation classification (AMC) in wireless communications. Unlike standard Masked Autoencoders (MAEs), which solely reconstruct masked inputs, DenoMAE2.0 jointly performs denoising and reconstruction by incorporating a position-aware local patch classification objective. This approach enables the model to simultaneously denoise corrupted signals and accurately reconstruct missing information, effectively capturing both global context and local structural patterns crucial for modulation classification under noisy and data-scarce scenarios. Extensive experiments demonstrate that DenoMAE2.0 achieves superior denoising and classification performance, outperforming baseline methods and its predecessor, DenoMAE, particularly under extremely low signal-to-noise (SNR) ratios (down to -7 dB), where other approaches significantly degrade. Our method achieves state-of-the-art accuracy of 82.4%, improving upon the original DenoMAE by 1.1%, while consistently enhancing reconstruction quality across all modulation classes. Additionally, DenoMAE2.0 exhibits robust transfer learning capabilities, achieving consistent improved accuracy over a vision transformer (ViT) at different SNRs with a 6.13% highest gain at noise-signal equilibrium point. These results highlight the effectiveness of our dual-objective, self-supervised framework for robust AMC in challenging wireless communication environments.

I. INTRODUCTION

Semi-supervised models have emerged as a transformative paradigm in machine learning, addressing the high costs associated with data labeling [1], [2]. A prevalent approach in semi-supervised learning involves pretraining models on large amounts of unlabeled data, which are abundant and readily available [3], [4]. These models leverage unlabeled data to capture underlying patterns and structures, enabling effective representation learning. Consequently, the pretrained models exhibit improved performance and faster adaptation during downstream tasks, making semi-supervised learning a cost-efficient and scalable solution for real-world applications [5].

Recent advancements in self-supervised learning have achieved remarkable success through representation learning [6], [7], contrastive learning [8], [9], and masking strategies [10]. Among these, masking strategies leveraging Vision Transformer (ViT) [11] architectures have set new benchmarks in tasks such as classification, reconstruction, and image analysis. Masked Autoencoders (MAE) [12], in particular, have demonstrated significant potential by randomly masking portions of input data and training models to reconstruct the missing information. Despite their success, traditional MAE methods predominantly emphasize global reconstruction objectives, often neglecting fine-grained local patterns that are critical for enhancing representation learning [13].

Masking strategies are particularly beneficial for pre-training in wireless communication for Automatic Modulation Classification (AMC), where acquiring labeled data is challenging due to privacy and copyright constraints [14], [15]. Moreover, communication signals are often corrupted by environmental noise and channel losses, making robust representation learning essential. DenoMAE [16] addresses these challenges by integrating masking and denoising within a unified framework, leveraging masked modeling to enhance performance. While DenoMAE has demonstrated exceptional results in denoising and achieving strong classification accuracy, it inherits the fundamental principles of MAE, which focus primarily on global representation learning through masking. Consequently, the model remains unaware of the specific locations of the masked inputs, limiting its ability to fully exploit fine-grained local patterns and unlock the complete potential of representation learning.

We present DenoMAE2.0, an enhanced framework that introduces a novel local patch classification objective alongside the traditional reconstruction task. Our approach differs from existing methods by treating visible patches as distinct classes based on their spatial positions, enabling the model to learn position-aware local features while maintaining global coherence. This

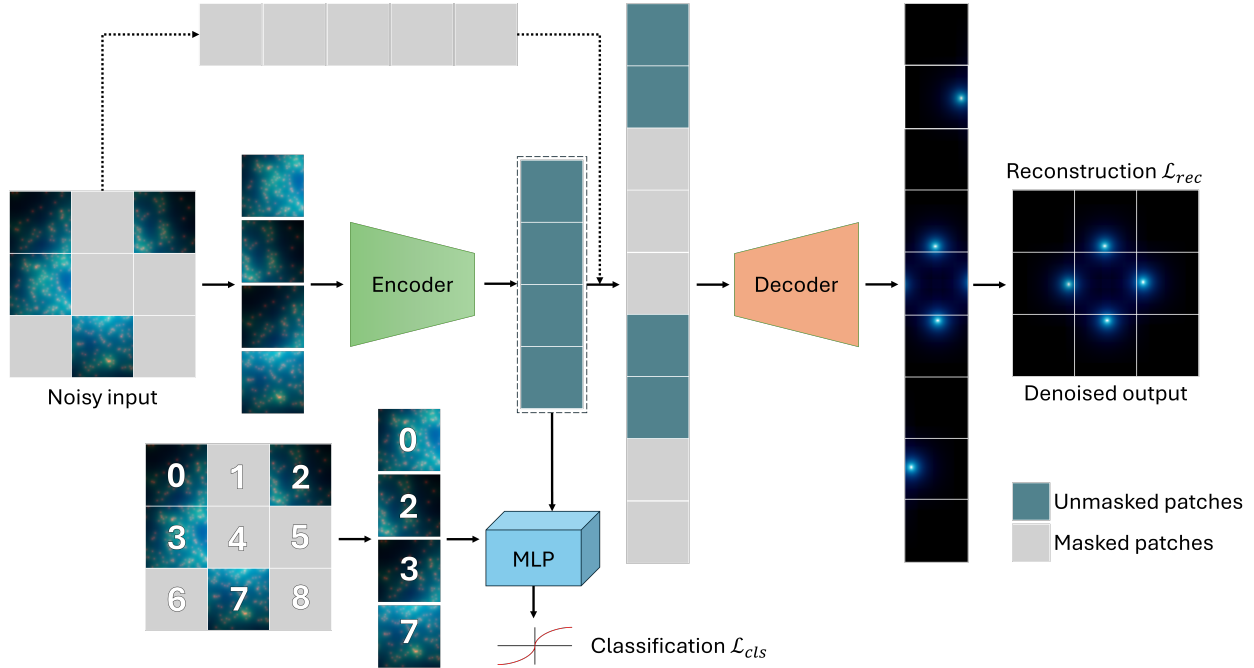


Fig. 1: DenoMAE2.0 architecture overview: The model consists of an encoder-decoder architecture. The encoder processes visible patches and mask tokens, producing patch-level features for reconstruction and classification. The decoder generates reconstructed patches, while the classification head predicts patch position based on visible patch features.

dual-objective strategy encourages the model to capture both structural and semantic information, leading to more robust and informative representations. The key contributions of our work are threefold:

- We propose a novel architecture that combines denoising reconstruction with local patch classification, enabling simultaneous learning of global and local features.
- We introduce a position-based classification strategy that leverages spatial information to enhance representation learning without requiring additional labels.
- Through extensive experiments, we demonstrate that DenoMAE2.0 achieves superior performance on various downstream and transfer learning tasks compared to traditional MAE and DenoMAE approaches, particularly in scenarios with limited labeled data.

Our results indicate that the integration of local patch classification significantly improves the quality of learned representations, leading to better denoising, generalization and faster convergence during fine-tuning. The proposed method not only advances the state-of-the-art in self-supervised learning but also provides insights into the importance of incorporating local structural information in representation learning frameworks.

The remainder of this paper is organized as follows: Section II reviews related work in self-supervised learning and automatic modulation classification. Section III details our proposed DenoMAE2.0 architecture and methodology. Section IV describes the experimental setup, datasets, and implementation details. Section V presents our results on denoising performance, classification accuracy, and transfer learning capabilities. Section VI provides an ablation study analyzing the contribution of individual components. Finally, Section VII concludes the paper.

II. RELATED WORK

A. AMC in Wireless Communication

Early AMC methods primarily relied on expert-crafted features extracted from constellation diagrams and signal statistics, utilizing likelihood-based approaches and higher-order statistics to classify modulation schemes [17]. Constellation diagrams, which visually represent symbol distributions in the complex plane, have been a fundamental tool for feature extraction in classical AMC. These diagrams illustrate the effects of noise, interference, and channel distortions, making them valuable for differentiating modulation schemes based on clustering patterns, symbol dispersion, and trajectory structures [18].

The advent of deep learning revolutionized AMC by enabling models such as CNNs to operate directly on I/Q samples and constellation diagrams, bypassing the need for manual feature engineering [19]. Deep learning approaches leverage spatial patterns in constellation diagrams, learning robust representations that improve classification accuracy over conventional methods [20]. More recent advancements incorporate attention mechanisms and transformer architectures to capture long-range dependencies in signal patterns [21]. However, these deep learning-based AMC methods often require large labeled datasets and exhibit limited generalization across varying channel conditions, posing challenges for real-world deployment [22].

B. Self-Supervised Learning Approaches

The evolution of modern Self-Supervised Learning (SSL) strategies in deep learning and wireless communication has progressed through distinct phases, starting with contrastive learning, followed by representation learning, and more recently, the adoption of masking techniques. Contrastive learning exemplified by SimCLR [8] and MoCo [23], relied on distinguishing positive and negative sample pairs to learn robust feature representations, proving effective in image and signal processing tasks. This was followed by representation learning methods [6], [24], which eliminated the need for negative samples by aligning representations of augmented views, enhancing efficiency and scalability. More recently, masking-based techniques [12], [25] have gained prominence by reconstructing masked portions of inputs, offering a powerful framework for learning from incomplete data [16].

In wireless communication, semi-supervised learning are particularly useful for modulation classification, where labeled data is often scarce due to the complexity of signal environments. Ma et al. [26] developed a semi-supervised framework leveraging pseudo-labeling, where a model trained on limited labeled modulation samples generates labels for unlabeled data, iteratively refining its performance across diverse modulation schemes. Another approach by Sifaou and Simeone [27] exploits Cross-Prediction-Powered Inference (CPPI) to enhance semi-supervised learning by mitigating the bias of synthetic labels. Their method refines ML-based predictions through tuned CPPI and meta-CPPI, achieving improved performance in tasks like beam alignment and indoor localization, particularly when labeled data is scarce. More related SSL methods in wireless communication includes convolution [28]–[30] and transformer [31]–[33] based SSLs.

III. METHODOLOGY: DENOMAE2.0

The overall framework of the proposed DenoMAE2.0 is illustrated in Figure 1. DenoMAE2.0 consists of three components: encoder, reconstruction denoising decoder, and local patch classification head. There are two complementary objectives: the denoising reconstruction objective and the local patch classification classification objective. Details are provided in Algorithm 1 and introduced as follows.

A. Masking Strategy and Encoder Architecture

Consider a pair of images: a noisy input image $\mathbf{x}_{\text{noisy}} \in \mathbb{R}^{H \times W \times C}$ and its corresponding clean target image $\mathbf{x}_{\text{clean}} \in \mathbb{R}^{H \times W \times C}$. The encoder processes the noisy image $\mathbf{x}_{\text{noisy}}$, while the decoder is trained to reconstruct the clean image $\mathbf{x}_{\text{clean}}$. We first partition the noisy input image $\mathbf{x}_{\text{noisy}}$ into

$$N = \frac{H W}{P^2} \quad (1)$$

non-overlapping patches, each of size $P \times P$. Let $\mathbf{x}_{p,\text{noisy}} \in \mathbb{R}^{N \times (P^2 C)}$ be the flattened noisy patches. Similarly, we partition the clean target image into patches $\mathbf{x}_{p,\text{clean}} \in \mathbb{R}^{N \times (P^2 C)}$. A linear projection E_{patch} maps the noisy patches into a d -dimensional embedding space:

$$\mathbf{z}_p = E_{\text{patch}}(\mathbf{x}_{p,\text{noisy}}) \in \mathbb{R}^{N \times d}. \quad (2)$$

To preserve spatial order, we add learnable positional embeddings $\mathbf{p} \in \mathbb{R}^{N \times d}$, forming the initial token sequence

$$\mathbf{Z}^{(0)} = \mathbf{z}_p + \mathbf{p}. \quad (3)$$

A random masking operator $\mathcal{M}_r$ discards a fraction r of these tokens to yield the visible tokens $\mathbf{Z}_v$. The masked tokens $\mathbf{Z}_m$ are replaced by mask tokens in the decoder but are omitted here in the encoder. Let $\mathcal{E}$ denote the ViT-based encoder with L Transformer layers. Each layer ℓ applies multi-head self-attention (MSA) and a feed-forward network (FFN) to the visible tokens:

$$\mathbf{Z}_v^{(\ell)} = \mathbf{Z}_v^{(\ell-1)} + \text{MSA}\left(\text{LN}(\mathbf{Z}_v^{(\ell-1)})\right), \quad (4)$$

$$\mathbf{Z}_v^{(\ell)} = \mathbf{Z}_v^{(\ell)} + \text{FFN}\left(\text{LN}(\mathbf{Z}_v^{(\ell)})\right). \quad (5)$$

After L layers, the encoder produces latent representations

$$\mathbf{q}_v = \mathcal{E}(\mathbf{Z}_v), \quad (6)$$

which are subsequently used for both reconstruction in the decoder and the auxiliary classification of unmasked patches.

B. Decoder Architecture and Reconstruction

To reconcile the difference between the visible token set $\mathbf{Z}_v$ (equation (6)) and the full patch set of size N (equation (1)), we introduce learnable *mask tokens* at the masked positions. These tokens, which share a single learned embedding, act as placeholders for the missing patches. Formally, let $\mathbf{M} \in \mathbb{R}^{|\mathcal{I}_m| \times d}$ be the mask token embeddings corresponding to the set of masked indices $\mathcal{I}_m$, where $\mathcal{I}_v \cup \mathcal{I}_m = \{1, 2, \dots, N\}$ and $\mathcal{I}_v \cap \mathcal{I}_m = \emptyset$. We concatenate the visible patch embeddings $\mathbf{q}_v$ and $\mathbf{M}$, then add positional embeddings to form the decoder input:

$$\mathbf{Z}_d^{(0)} = \text{Shuffle}(\mathbf{q}_v \parallel \mathbf{M}) + \mathbf{p}_d, \quad (7)$$

where $\mathbf{p}_d \in \mathbb{R}^{N \times d}$ denotes the positional embeddings for the decoder, and $\text{Shuffle}(\cdot)$ reorders the tokens to match their original patch indices.

The decoder consists of a Transformer stack with fewer layers and heads than the encoder for computational efficiency. Denote the decoder by $\mathcal{D}$. Each layer follows the same multi-head self-attention and feed-forward structure described in (4)–(5), yielding a final decoder output $\mathbf{Z}_d^{(L')}$. A linear projection $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times (P^2 C)}$ maps each token to the original patch dimension, thereby producing reconstructed patches:

$$\hat{\mathbf{x}} = [\mathbf{Z}_d^{(L')} \mathbf{W}_{\text{out}}] \in \mathbb{R}^{N \times (P^2 C)}. \quad (8)$$

We adopt a *masked* mean squared error (MSE) loss to focus the reconstruction objective solely on the masked patches. Let $\hat{\mathbf{x}}_m$ denote the reconstructed patches and $\mathbf{x}_{m,\text{clean}}$ denote the corresponding clean target patches for the set of masked indices $\mathcal{I}_m$. The reconstruction loss is

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{I}_m|} \sum_{i \in \mathcal{I}_m} \|\hat{\mathbf{x}}_m^{(i)} - \mathbf{x}_{m,\text{clean}}^{(i)}\|^2, \quad (9)$$

where $\|\cdot\|$ denotes the Euclidean norm. By confining $\mathcal{L}_{\text{rec}}$ to the masked regions and comparing against clean target patches, the decoder is compelled to infer the missing patches from noisy context while learning to denoise, thus enhancing its denoising capabilities.

C. Patch Classification Head

While the decoder focuses on reconstructing masked patches from clean targets, the visible patch embeddings $\mathbf{q}_v$ from the noisy input (equation (6)) can also facilitate a classification objective. Unlike traditional classification tasks where a single label is assigned per image, we treat *each unmasked patch* as a distinct class. Let $\mathbf{q}_v \in \mathbb{R}^{|\mathcal{I}_v| \times d}$ represent the visible token embeddings

corresponding to the set of indices $\mathcal{I}_v$. We define the total number of classes by

$$N_{\text{cls}} = N \cdot (1 - r), \quad (10)$$

where N is the total number of patches (equation (1)), and r is the masking ratio. Hence, each visible patch is assigned a unique position-based label.

To learn these labels, we project the visible embeddings $\mathbf{q}_v$ via a simple linear layer $\mathbf{W}_{\text{cls}} \in \mathbb{R}^{d \times N_{\text{cls}}}$, yielding logits $\mathbf{p} \in \mathbb{R}^{|\mathcal{I}_v| \times N_{\text{cls}}}$. The classification loss $\mathcal{L}_{\text{cls}}$ is defined via cross-entropy:

$$\mathcal{L}_{\text{cls}} = \text{CE}(\mathbf{p}, \mathbf{y}), \quad (11)$$

where $\mathbf{y}$ denotes the ground-truth position labels for the visible patches. By jointly optimizing (9) and (11), the model learns richer representations of both global context and local patch identities.

D. Overall Objective

DenoMAE2.0 optimizes a combination of reconstruction and classification losses, enabling simultaneous learning of fine-grained local and global features. The overall loss function is formulated as a weighted sum:

$$\mathcal{L} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} \quad (12)$$

where λ_{rec} and λ_{cls} are balancing weights for the reconstruction and classification objectives, respectively. Through empirical validation (see Table IX), we set $\lambda_{\text{rec}} = 1.0$ and $\lambda_{\text{cls}} = 0.1$.

E. Fine-Tuning for Classification Tasks

After pre-training DenoMAE2.0 on unlabeled data, we fine-tune the model for downstream classification. First, we discard the decoder and the patch classification head, retaining only the encoder (equation (6)). During fine-tuning, we provide the encoder with full, unmasked images, thereby omitting the masking mechanism. Let $\{\mathbf{z}_i\}_{i=1}^N$ denote the patch embeddings produced by the encoder for each input, where N remains as defined in (1). We apply mean pooling:

$$\mathbf{z} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i, \quad (13)$$

to obtain a global image representation.

This $\mathbf{z}$ is then passed through a lightweight multi-layer perceptron (MLP) head, defined as:

$$f_{\text{cls}}(\mathbf{z}) = W_3 \cdot \text{GELU}(W_2 \cdot \text{GELU}(W_1 \cdot \mathbf{z})) \quad (14)$$

where $\mathbf{z} \in \mathbb{R}^{d_{\text{encoder}}}$ is the image representation, $W_1 \in \mathbb{R}^{512 \times d_{\text{encoder}}}$, $W_2 \in \mathbb{R}^{256 \times 512}$, and $W_3 \in \mathbb{R}^{C \times 256}$ with C denoting the number of classes in the dataset. Dropout

with rate $p = 0.1$ is applied after each GELU activation [34].

This representation is subsequently passed through the classification head to produce final predictions $\mathbf{y} = f_{\text{cls}}(\mathbf{z})$. By default, we maintain frozen encoder parameters during fine-tuning to preserve pre-trained representations, although this constraint can be relaxed for end-to-end optimization when sufficient labeled data is available.

IV. DATASETS

A. Constellation Diagram Generation

For modulation classification tasks, constellation diagrams provide superior insights compared to raw time-series data by capturing more detailed signal characteristics [20]. The constellation diagram generation process involves several key steps. Initially, modulated signals are projected onto a complex plane, balancing detail preservation with computational efficiency for SNR levels between -10 dB and 10 dB. The visualization process then evolves through multiple phases: beginning with a basic gray-scale representation that handles varying sample densities per pixel, followed by an enhanced version incorporating an exponential decay function ($B_{i,j}$). This enhancement accounts for both the precise sample locations within pixels and their neighborhood effects, considering factors like sample power (P), sample-to-pixel centroid distances ($d_{i,j}$), and decay rate (α) [35]. To align with DenoMAE2.0's input requirements, we create a three-channel RGB image by generating three separate enhanced grayscale images from identical samples, each using different decay parameters.

Our dataset includes ten modulation types as outlined in Table I. The pretraining dataset comprises 10,000 samples uniformly distributed across ten modulation types, with randomly assigned SNR values in the range of -10 dB to 10 dB. For the downstream classification task, we use 1000 samples for training and 100 samples for testing, evenly distributed among the 10 classes, evaluated at SNR values from -10 dB to 10 dB in 1 dB increments.

TABLE I: List of all modulation classes

Modulation Technique	Abbreviation
4-Amplitude Shift Keying	4ASK
4-Pulse Amplitude Modulation	4PAM
8-Amplitude Shift Keying	8ASK
16-Pulse Amplitude Modulation	16PAM
Continuous-Phase Frequency Shift Keying	CPFSK
Differential Quadrature Phase Shift Keying	DQPSK
Gaussian Frequency Shift Keying	GFSK
Gaussian Minimum Shift Keying	GMSK
On-Off Keying	OOK
Offset Quadrature Phase Shift Keying	OQPSK

Algorithm 1 DenoMAE2.0, PyTorch-like pseudo code

```

# f_enc, f_dec: encoder, decoder
# f_mlp: patch classification head
# mask_r: mask ratio
# lambda_rec, lambda_cls: loss function weights

1: # load noisy-clean pairs
2: for x_noisy, x_clean in loader do
3:   # noisy patch embeddings
4:   x_noisy = patch_emb(x_noisy)
5:   # clean patch embeddings
6:   x_clean = patch_emb(x_clean)
7:   # random visible and masked patches from noisy input
8:   x_v, x_m_noisy, mask_ids =
       masking(x_noisy, mask_r=mask_r)
9:   # same mask for clean
10:  _, x_m_clean, _ =
       masking(x_clean, mask_ids=mask_ids)
11:  # unmasked noisy patch features
12:  q_v = f_enc(x_v)
13:  logits = f_mlp(q_v)  ▷ predicted patch labels
14:  k_m = f_dec(q_v)     ▷ pixels reconstruction

15:  # class labels generation
16:  labels = class_labels(mask_ids)
17:  # compare with clean
18:  loss = lambda_rec *
         rec_loss(k_m, x_m_clean) +
         lambda_cls *
         cls_loss(logits, labels)
19:  loss.backward()
20:  optimizer.update()  ▷ optimizer update
21: end for

# label generalization for patches
function class_labels(mask_ids):
22:  classes = ((img_size//patch_size)2) *
             (1 - mask_r)
23:  labels =  ▷ 0: visible, 1: masked
             classes[mask, where mask == 0]
24:  return labels

function rec_loss(k, x):  ▷ reconstruction loss
25:  x = normalize(x)      ▷ normalize patches
26:  loss = (k - x)2       ▷ MSE loss
27:  return loss

function cls_loss(logits, tgt):  ▷ classification loss
28:  loss = CrossEntropyLoss(logits, tgt)
29:  return loss

```

Note: ‘patch_emb’ and ‘masking’ functions are omitted for brevity. For full implementation, see: <https://github.com/atik666/denoMAE2>.

B. Parameters in Sample Generation

Our dataset incorporates ten distinct modulation formats. Each signal initially has a length of $L_0 = 1024$ samples. The preprocessing pipeline adapts these signals for transformer architecture compatibility through two main steps: first transforming into $\mathbf{S}_1 \in \mathbb{R}^{32 \times 32}$, then interpolating to $\mathbf{S}_2 \in \mathbb{R}^{224 \times 224}$. For modulation formats exceeding L_0 (such as GMSK with $L_{\text{GMSK}} = 8196$), we perform initial downsampling to L_0 . The final signal representation $\mathbf{S}_3 \in \mathbb{R}^{3 \times 224 \times 224}$ is created by triplicating $\mathbf{S}_2$ to match the three-channel structure of constellation images $\mathbf{I} \in \mathbb{R}^{3 \times 224 \times 224}$. A uniform sampling rate of $f_s = 200$ kHz is maintained across all modulation formats.

C. Model Parameters and Hyperparameters

DenoMAE2.0 architecture uses distinct configurations for pretraining and fine-tuning phases (as shown in Table II and Table III). The model processes 224×224 pixel images divided into 16×16 patches. During pretraining, we employ an encoder-decoder structure where the encoder features 12 layers with 12 attention heads and an embedding dimension of 768, while the decoder is lighter with 8 layers, 8 attention heads, and a 512-dimensional embedding space. A mask ratio of 0.75 is applied during the self-supervised pretraining phase, balancing reconstruction difficulty with information preservation.

For fine-tuning, only the encoder portion is retained with its 12-layer, 12-attention-head configuration maintained, adapting the output for either 10-class (original dataset) or 24-class (extended dataset) classification tasks. The learning rate is reduced from 0.0003 during pretraining to 0.0001 for fine-tuning, reflecting the transition from broad feature learning to targeted classification optimization. Both phases maintain a weight decay of 0.05 for regularization, but batch sizes differ (64 for pretraining vs. 32 for fine-tuning) to accommodate the computational demands of each training regime.

V. EXPERIMENTAL RESULTS

In this section, we evaluate DenoMAE2.0's denoising capabilities, latent space representation quality, downstream classification accuracy, and robustness across different SNRs. The results demonstrate consistent improvements over baseline methods and the original DenoMAE architecture.

A. Denoising Performance

During the pretraining phase, DenoMAE2.0 learns to denoise and reconstruct missing patches, thereby developing robust signal representations. We evaluate this denoising and reconstruction performance both during and

TABLE II: List of hyperparameters during pretraining

Parameter name	Value
Training batch size	64
Testing batch size	10
Image size	224, 224
Patch size	16
Mask ratio	0.75
Encoder embedding dimension	768
Decoder embedding dimension	512
Encoder depth	12
Decoder depth	8
Encoder number of heads	12
Decoder number of heads	8
Number of epochs	100
Learning rate	0.0003
Weight decay	0.05
Classification loss weight	0.1
Reconstruction loss weight	1.0

TABLE III: List of hyperparameters during fine-tuning

Parameter name	Value
Training batch size	32
Testing batch size	10
Image size	224, 224
Patch size	16
Encoder embedding dimension	768
Encoder depth	12
Encoder number of heads	12
Number of epochs	150
Learning rate	0.0001
Weight decay	0.05
Number of classes	10

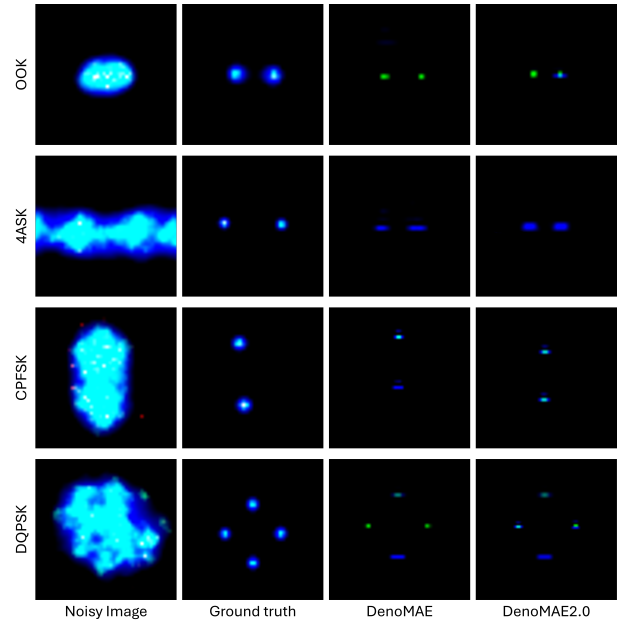


Fig. 2: Reconstruction performance comparison between DenoMAE and DenoMAE2.0

TABLE IV: Quantitative Comparison of Denoising Performance between DenoMAE and DenoMAE2.0.

Class	SSIM		PSNR	
	DenoMAE	DenoMAE2.0	DenoMAE	DenoMAE2.0
16PAM	0.9422	0.9464	41.6867	41.8003
4ASK	0.9607	0.9634	42.1780	42.3096
4PAM	0.9328	0.9643	42.0005	42.1061
8ASK	0.9745	0.9813	42.8006	42.9158
CPFSK	0.9719	0.9738	42.3492	42.5520
DQPSK	0.9356	0.9360	41.8405	41.9223
GFSK	0.9677	0.9681	42.0733	42.1021
GMSK	0.8749	0.8771	39.9367	40.2160
OOK	0.9586	0.9674	42.3674	42.6585
OQPSK	0.9129	0.9204	41.0052	41.1172

after pretraining. Figure 2 illustrates the effectiveness of DenoMAE2.0 through reconstructed constellation diagrams compared to the original DenoMAE model. DenoMAE2.0 achieves superior noise reduction while preserving critical signal features necessary for accurate modulation classification.

Table IV quantifies these improvements using two standard image quality metrics: structural similarity index (SSIM) and peak signal-to-noise ratio (PSNR). We report the mean SSIM and PSNR scores across all validation images (ten images per class) during pretraining. The results demonstrate that DenoMAE2.0 consistently outperforms the original DenoMAE across all modulation classes. For SSIM, DenoMAE2.0 shows improvements ranging from minor enhancements (0.0004 increase for DQPSK) to substantial gains (0.0315 increase for 4PAM). Similarly, for PSNR, DenoMAE2.0 achieves consistent improvements across all classes, with particularly notable enhancements for GMSK (0.2793 dB increase) and OOK (0.2911 dB increase). These consistent improvements in reconstruction quality metrics directly contribute to the enhanced downstream classification performance of DenoMAE2.0.

B. Latent Space Visualization with TSNE

In Figure 3, we visualize the latent representations of DenoMAE and DenoMAE2.0 using t-distributed stochastic neighbor embedding (t-SNE) [36]. The plots (left two) reveal that DenoMAE2.0 produces more accurate and distinct clusters than DenoMAE, indicating superior feature separation and representation learning. Specifically, in DenoMAE, classes 0, 3, and 9 contain more separated clusters within the same class. In contrast, DenoMAE2.0 exhibits more distinct clusters for these classes.

This enhanced clustering is particularly evident in the masked samples (right two plots), where we mask out a significant portion (75%) of the images. Masking typically makes distinct cluster formation more challenging. However, DenoMAE2.0 achieves clearer boundaries

between classes, suggesting improved robustness and generalization capabilities. In DenoMAE, classes 0, 3, and 9 achieve somewhat distinct clusters, with a few samples overlapping in other clusters. In DenoMAE2.0, classes 0, 3, 6, 7, 8, and 9 show better-separated clusters, indicating enhanced feature learning and representation.

However, in all the plots, classes 1 and 2 entirely overlap with each other, indicating that the model struggles to distinguish between these two classes. This is a common issue in modulation classification, as some classes are very similar to each other.

C. Finetuning Accuracy

After pretraining, we transfer the learned representations to the downstream classification task and compare DenoMAE2.0 with state-of-the-art models. As shown in Table V, when finetuned on the downstream task, DenoMAE2.0 achieves the highest test accuracy of 82.40% among all compared methods. This represents a significant improvement over the baseline ViT (79.90%) and other representation learning approaches including DEiT (81.20%), MoCov3 (81.00%), BEiT (80.40%), MAE (80.10%), and the original DenoMAE (81.30%). The consistent performance advantage demonstrates that DenoMAE2.0's enhanced denoising strategy learns more effective representations during pretraining that transfers well to the downstream classification task.

TABLE V: Downstream classification accuracy for different models

Method	Test accuracy (%)
ViT [11]	79.90
DEiT [37]	81.20
MoCov3 [38]	81.00
BEiT [39]	80.40
MAE [12]	80.10
DenoMAE [16]	81.30
DenoMAE2.0 (Ours)	82.40

D. Confusion Matrix

Figure 4 presents confusion matrices visualizing the per-class performance of both DenoMAE and Deno-

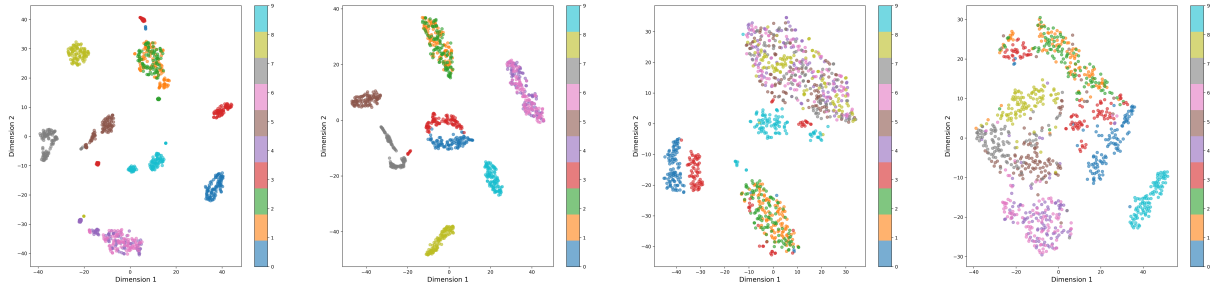


Fig. 3: Latent representation visualization using t-SNE. From left to right: (1) DenoMAE without masking, (2) DenoMAE2.0 without masking, (3) DenoMAE with 75% masking, and (4) DenoMAE2.0 with 75% masking.

MAE2.0. The diagonal elements indicate correct classifications, while off-diagonal elements show misclassifications. Both models demonstrate perfect classification accuracy (100%) for several modulation types, specifically 16PAM, 4ASK, 8ASK, DQPSK, GMSK, OOK, and OQPSK.

However, persistent classification challenges are evident between 4ASK and 4PAM, where both models show significant confusion with 50% accuracy for each class and 50% misclassification between them. This symmetric confusion pattern suggests inherent signal similarities between these amplitude modulation schemes that remain difficult to distinguish even with our enhanced architecture. Notably, this confusion pattern directly aligns with observations from the t-SNE analysis, where classes 1 and 2 show complete overlap in the latent space visualization, further confirming the inherent similarity in signal characteristics between these modulation types.

Additionally, both models struggle with differentiating between CPFSK and GFSK, though DenoMAE2.0 shows modest improvement in CPFSK recognition (70% accuracy compared to 63% in DenoMAE). Despite this improvement, GFSK classification remains challenging with DenoMAE2.0 achieving only 54% accuracy (compared to 52% in DenoMAE), with most misclassifications assigning GFSK signals to the CPFSK class. This confusion pattern aligns with the theoretical similarity between these frequency shift keying techniques, as GFSK is essentially CPFSK with additional Gaussian filtering. These consistent patterns across both model versions indicate fundamental signal similarities that present intrinsic classification challenges rather than model limitations.

E. Performance on Number of Epochs

Figure 5 illustrates the finetuning performance across different epochs. The plot shows a steady improvement in model accuracy as the number of epochs increases, with DenoMAE2.0 consistently outperforming baseline

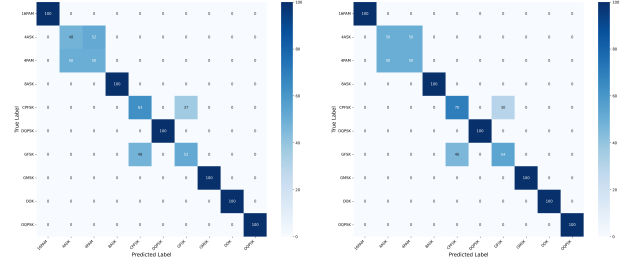


Fig. 4: Confusion matrices for downstream classification. Left: DenoMAE, Right: DenoMAE2.0

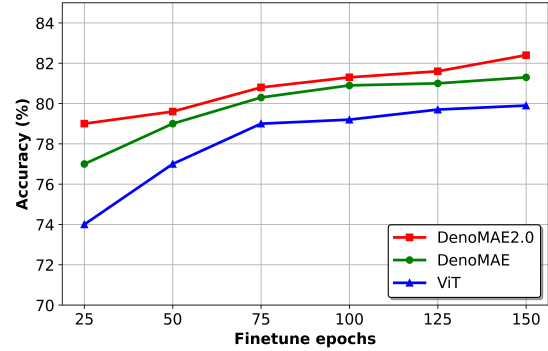


Fig. 5: Finetuning performance improvement for different epochs

methods. DenoMAE achieves higher accuracy than ViT continuously. On the other hand, although the gap between DenoMAE and ViT is larger than DenoMAE and DenoMAE2.0, DenoMAE2.0 always obtains better accuracy than DenoMAE.

F. Performance Across Different SNRs

The SNR performance plot demonstrates robust classification accuracy across various SNRs ranging from -10 dB to 10 dB. As the SNR values increases all the methods perform better and when the SNR is low the performance degrades. ViT, which has no pertaining

phase, presumably obtains the lowest performance in all cases. MAE, which pretrains on modulation signals but has no denoising capability, obtains a slight higher accuracy than ViT. DenoMAE obtains higher accuracy than MAE in most cases however for SNE of 2 and 3 we observe that MAE obtains a higher accuracy. On the other hand, DenoMAE2.0 outperforms all the methods for all the SNRs. Noteably, all the methods lose significant accuracy for snr lower than -2 dB. However, DenoMAE2.0 maintains comparatively robust performance down to -7 dB, demonstrating its enhanced resilience to noise.

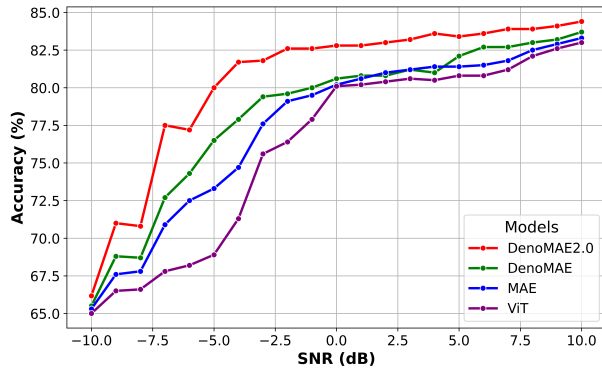


Fig. 6: Accuracy comparison across different SNRs

VI. COMPARISON WITH OTHER AMC METHODS

Table VI presents a comprehensive comparison between DenoMAE2.0 and other state-of-the-art modulation classification methods. AlexNet achieved 82.8% accuracy with 8 classes at 2dB SNR using a large dataset of 800,000 samples. NMformer reported 71.6% accuracy across 10 classes with SNR ranging from 0.5 to 4.5dB using 106,800 samples. CNN-AMC and DL-GRF showed relatively lower performance with 50.40% and 59% accuracy respectively, though they were tested under different conditions. The original DenoMAE achieved 81.3% accuracy across 10 classes with SNR ranging from 0 to 10dB. Our proposed DenoMAE2.0 demonstrates superior performance with 82.4% accuracy under similar conditions, using only 10,000 samples for pretraining and 1,000 samples for fine-tuning, indicating improved efficiency in both performance and data utilization.

VII. TRANSFERABILITY

To show the transferability of DenoMAE2.0 on other similar task, we performed experiments on a similar benchmark dataset named RadioML. A brief description on the RadioML dataset is provided in the following subsection.

A. RadioML Dataset

We use a subset of the RadioML 2018.01A dataset [18], which is widely used as a benchmark in wireless communication [43] and spectrum sensing research [44], facilitates advancements in deep learning-based modulation recognition, interference mitigation, and cognitive radio applications [45]–[47]. The dataset consists of I/Q (in-phase and quadrature) signal samples, each with shape 1024×2 . To evaluate performance across a comprehensive range of signal conditions, we selected five strategically chosen SNR levels: -20 dB, -10 dB, 0 dB, 10 dB, and 20 dB. This selection enables both interpolation analysis (at 0 dB and 10 dB, within DenoMAE2.0's pretraining range) and extrapolation assessment (at -20 dB, -10 dB, and 20 dB), providing insights into the model's generalization capabilities beyond its training distribution. For each modulation-SNR combination, we utilized 4,000 training samples, 46 validation samples, and 50 testing samples, balancing statistical significance with computational efficiency. In our study, we focus on the normal modulation types of RadioML dataset which consists of 11 classes as listed in Table VII.

B. Performance

To evaluate transferability, we applied our models pretrained on the original dataset to the RadioML benchmark through fine-tuning. This process tests whether the representations learned during pretraining generalize effectively to new signal distributions and classification tasks. Figure 7 illustrates classification accuracy across multiple SNR levels, demonstrating how transfer learning performance scales with signal quality.

At extremely low SNRs (-20 dB), all three methods converge to the same baseline performance (9.09%), indicating they reach a fundamental classification limit where signal features become indistinguishable from noise. This represents random guessing for an 11-class problem, suggesting that no model can extract meaningful patterns at this noise level. At moderately low SNR (-10 dB), DenoMAE2.0 begins to demonstrate its superior noise robustness by achieving 10.12% accuracy while both DenoMAE and ViT remain at baseline performance. This small but notable improvement (1.03 percentage points) represents the earliest evidence of DenoMAE2.0's enhanced feature extraction capabilities in highly degraded signals.

The most performance differentiation occurs at 0 dB SNR, where DenoMAE2.0 achieves 39.53% accuracy, outperforming DenoMAE by 3.2 percentage points and ViT by 6.13 percentage points. This substantial gap at the noise-signal equilibrium point highlights DenoMAE2.0's effectiveness precisely where denoising capa-

TABLE VI: Comparison of DenoMAE2.0 with other modulation classification methods

Methods	Number of samples		SNR dB	Number of classes	Finetuning parameters	Inference time (ms)	Test accuracy (%)
	Pretraining	Fine-tuning					
AlexNet [35]	800,000 (train)	8,000 (test)	2	8	22.5M	2.0	82.8
NMformer [40]	106,800	3,000	0.5 - 4.5	10	85.8M	14.5	71.6
CNN-AMC [41]	79,200	228,060	-6	4	1.6M	0.13	50.40
DL-GRF [42]	2,000 (train)	200 (test)	0	4	23.5M	10	59
DenoMAE [16]	10,000	1,000	0 - 10	10	85.8M	14.5	81.3
DenoMAE2.0 (Ours)	10,000	1,000	0 - 10	10	85.8M	14.5	82.4

TABLE VII: Modulation classes in the RadioML

Class Name	Abbreviation
4-level Amplitude Shift Keying	4ASK
8-phase Phase Shift Keying	8PSK
16-point Quadrature Amplitude Modulation	16QAM
Amplitude Modulation, Double Sideband Suppressed Carrier	AM-DSB-SC
Amplitude Modulation, Single Sideband Suppressed Carrier	AM-SSB-SC
Binary Phase Shift Keying	BPSK
Frequency Modulation	FM
Gaussian Frequency Shift Keying	GFSK
Offset Quadrature Phase Shift Keying	OQPSK
On-Off Keying	OOK
Quadrature Phase Shift Keying	QPSK

bilities matter most. In favorable SNR conditions (10-20 dB), all models achieve high accuracy (>90%), with DenoMAE2.0 maintaining its lead but with narrowing margins. The relative improvements become more modest (approximately 1 percentage point at 10 dB and 0.79 percentage points at 20 dB compared to DenoMAE), suggesting that when signals are cleaner, sophisticated denoising techniques offer diminishing returns.

The consistent hierarchy of performance (DenoMAE2.0 > DenoMAE > ViT) across the entire operational SNR range confirms that the architectural enhancements in DenoMAE2.0 provide systematic benefits for signal classification. Most importantly, the performance advantage is proportionally largest when signal quality is poor, precisely the condition where robust performance is most valuable in practical applications.

VIII. ABLATION STUDY

In this section, we analyze the impact of various components on the performance of our model.

A. Effect of the Loss Function

We conducted experiments to evaluate the contribution of different loss components in our model. As shown in Table VIII, we analyzed three configurations: using only reconstruction loss, only classification loss, and combining both losses. The results demonstrate

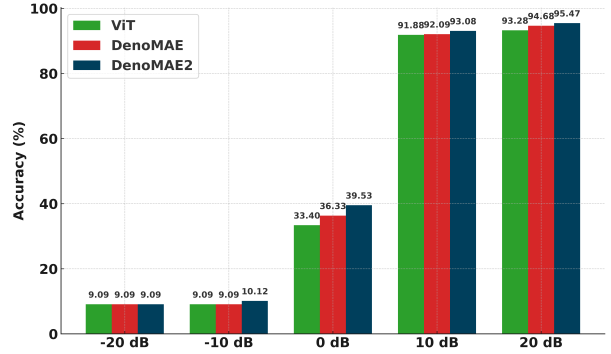


Fig. 7: Transfer learning performance on RadioML dataset

that while reconstruction loss alone achieves reasonable performance (81.30%), classification loss by itself underperforms (69.80%). However, combining both losses yields the best results (82.40%), suggesting that the joint optimization of reconstruction and classification objectives leads to more robust feature learning and better overall performance.

TABLE VIII: Accuracy for different loss configurations

Reconstruction Loss	Classification Loss	Accuracy (%)
✓		81.30
	✓	69.80
✓	✓	82.40

B. Effect of Auxiliary Loss Weights

We conducted experiments to analyze the impact of auxiliary loss weights on model performance, as shown in Table IX. The results demonstrate that the model's accuracy varies with different weight values, with 0.10 achieving the optimal performance at 82.4%. We observed a clear trend where very high weights (1.0) significantly degraded performance to 79.60%, while moderate weights (0.25, 0.50) maintained consistent performance at 82.0%. Lower weights (0.01, 0.05) showed slightly reduced performance at 81.3% and 81.5% respectively, indicating that the auxiliary loss weight needs to be

carefully tuned to achieve the best balance between learning objectives.

TABLE IX: Accuracy for different auxiliary loss weights

Aux. loss weights	0.01	0.05	0.10	0.25	0.50	1.0
Accuracy (%)	81.3	81.5	82.4	82.0	82.0	79.60

C. Combination of Losses

We investigated the effect of different weight combinations between reconstruction and auxiliary losses, as shown in Table X. The results reveal that balancing these two losses is crucial for optimal performance. Equal weighting (0.5, 0.5) yields suboptimal performance at 78.80%, while gradually increasing the reconstruction loss weight and decreasing the auxiliary loss weight shows steady improvement. The optimal combination is achieved with reconstruction loss weight of 1.0 and auxiliary loss weight of 0.10, resulting in 82.40% accuracy. Further increasing the reconstruction loss weight to 1.2 while reducing auxiliary loss weight to 0.05 leads to a slight performance degradation (82.0%), suggesting that maintaining a proper balance between these losses is essential for optimal model performance.

TABLE X: Accuracy for different combination of reconstruction and auxiliary loss weights

Rec. loss Weights	0.5	0.75	0.80	0.90	1.0	1.2
Aux. loss weights	0.5	0.25	0.10	0.10	0.10	0.05
Accuracy (%)	78.80	80.50	81.60	82.0	82.40	82.0

D. Effect of Number of MLP Hidden Dimensions

We examined the impact of varying MLP hidden dimensions on model performance, with results presented in Table XI. The experiments demonstrate that model performance improves as the hidden dimension size increases from 256 to 768, reaching peak accuracy at 768 dimensions (82.40%). Further increases to 1024 maintain this performance level, with a marginal improvement to 82.50% at 1280 dimensions. This suggests that while larger hidden dimensions can capture more complex features, the benefits plateau around 768-1024 dimensions, indicating this range as the optimal choice for balancing model capacity and computational efficiency.

TABLE XI: Accuracy results for different MLP hidden dimensions

MLP dimensions	256	512	768	1024	1280
Accuracy (%)	81.00	81.60	82.40	82.40	82.50

E. Effect of number of decoders

We investigated the impact of varying the number of decoders on model performance, as shown in Table XII. The results reveal a clear pattern where using a single decoder yields the lowest accuracy of 78.60%. A substantial improvement is observed when increasing to 3-4 decoders, both achieving 81.60%. Further enhancement is achieved with 8 decoders, reaching 82.40%, which plateaus through 12 decoders. Adding more decoders up to 16 provides only marginal improvement (82.50%). These results indicate that while multiple decoders are crucial for better feature learning, the performance gains saturate beyond 8 decoders, suggesting this as an optimal choice considering the computational efficiency trade-off.

TABLE XII: Accuracy results for different decoder sizes

No. of decoders	1	3	4	8	12	16
Accuracy (%)	78.60	81.60	81.60	82.40	82.40	82.50

IX. CONCLUSION

In this work, we presented DenoMAE2.0, an enhanced denoising autoencoder that jointly optimizes reconstruction loss and an unmasked patch classification loss to improve robustness and performance. By incorporating a classification head that refines feature learning on unmasked regions, DenoMAE2.0 surpasses its predecessor, DenoMAE, achieving state-of-the-art results across multiple benchmarks. Extensive experiments demonstrate its superior ability to handle noisy data while preserving high classification accuracy. Additionally, we investigate the influence of auxiliary loss weighting and MLP hidden dimensions on performance, offering key insights into the role of hyperparameter tuning in optimizing denoising autoencoders. Furthermore, the transfer learning evaluations conducted on the RadioML dataset confirm the superior generalization capabilities of our model across different modulation schemes and varied signal environments. As a result, DenoMAE2.0 presents an efficient and highly effective solution for modulation classification challenges in noisy communication scenarios, showcasing strong potential for real-world wireless communication deployments.

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